

INVISIBLE BUT DEADLY

INDOOR RADON EXPOSURE IN ALBERTA HOMES



A preventable environmental health risk affecting thousands of Albertans

WHAT IS RADON?

Radon is a colorless, odorless radioactive gas that naturally forms from the decay of uranium in soil (Health Canada, 2023). It can move through cracks and openings in the ground and accumulate inside homes, especially in basements (Health Canada, 2019). Radon is the second leading cause of lung cancer in Canada, and Alberta consistently reports some of the highest indoor radon levels in the country (Health Canada, 2023; WHO, 2021).



- **1 IN ALBERTA HOMES EXCEED HEALTH CANADA'S 200 Bq/m³ GUIDELINE (Health Canada, 2019)**
- **HIGHEST LEVELS SEEN IN CALGARY, EDMONTON, LETHBRIDGE, RED DEER (STANLEY ET AL., 2021)**
- **COLD CLIMATE – TIGHTLY SEALED HOMES – HIGHER ACCUMULATION (NCCEH, 2021)**

WHO IS AFFECTED?

- Children, seniors, and individuals living long term in basements (NCCEH, 2021; Health Canada, 2023)
- Low income renters with limited testing or mitigation options
- Rural and remote communities with reduced access to test kits
- New high efficiency (airtight) homes that trap radon more easily

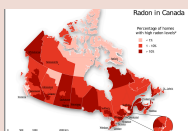


FIG: MAP OF AB WITH HIGH RISK REGIONS SHADED

EVIDENCE- WHAT WE KNOW

- Health Canada estimates 3,300+ lung cancer deaths per year from radon (Health Canada, 2023)
- 33% of Alberta homes exceed WHO's recommended level (100 Bq/m³) (Stanley et al., 2021) (WHO, 2021)
- Long term radon exposure increases lung cancer risk 10x for smokers (Health Canada, 2019; Darby et al., 2005)



WHY IT'S A POLICY PROBLEM?



- 1.Radon testing is voluntary in Alberta (NCCEH, 2021; Health Canada, 2019)
- 2.No mandatory disclosure for home buyers or renters
- 3.Low income households face financial barriers to mitigation
- 4.School, daycares, and public buildings lack consistent testing requirements (NCCEH, 2021)

EVIDENCE- BASED SOLUTION

Provincial Radon Mitigation & Disclosure Program (RMPD):

- Mandatory radon testing during home sale or rental
- Subsidized long term radon test kits for high risk and low income communities
- \$500 provincial rebate for radon mitigation systems
- Province wide public education campaign via Alberta Health Services
- School and daycare radon testing requirements

(Health Canada, 2019; Stanley et al., 2021)

Why this is feasible in Alberta: Typical mitigation cost is \$2,500 one time (Health Canada, 2019). Moreover high public support demonstrated in previous AB pilot programs and successful implementation examples in BC, Ontario and several U.S states (NCCEH, 2021).

DATA VISUALIZATION:

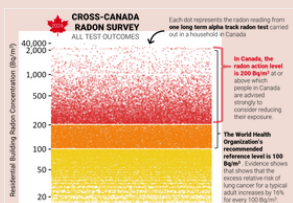


FIGURE 1. PROPORTION OF HOMES ABOVE RADON GUIDELINE LEVELS BY PROVINCE (CROSS-CANADA RADON SURVEY, 2024) (HEALTH CANADA, 2019) (CROSS-CANADA RADON SURVEY, 2024) (STANLEY ET AL., 2021)

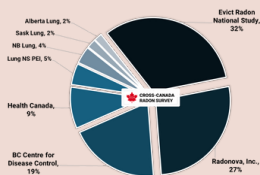


FIGURE 2. DISTRIBUTION OF RADON TEST RESULTS IN CANADIAN HOUSEHOLDS (HEALTH CANADA, 2024) (HEALTH CANADA, 2019) (CROSS-CANADA RADON SURVEY, 2024) (STANLEY ET AL., 2021)